Laboratory 05: Generative Models for Classification

Pattern Recognition

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Abstract

This report details the implementation, evaluation, and analysis of Generative Models (Standard MVG, Tied Covariance MVG, and Naive Bayes) applied to the project dataset. It explores the validity of the Gaussian assumption, the correlation between features, and the effects of feature ablation and PCA dimensionality reduction.

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1 Baseline model evaluation

Apply the MVG model to the project data. Compute the corresponding error rate.

Running the Standard Multivariate Gaussian (MVG) model on the full 6-dimensional dataset yields an accuracy of **93.0%**, which corresponds to an error rate of **7.0%**.

The standard MVG model achieves this strong performance because it computes a full, class-specific covariance matrix. This allows the model to capture both the individual variance of each feature and the covariances between them, independently for the "Fake" and "Genuine" classes. As observed in Laboratory 4, features 1, 2, 3, and 4 are reasonably well-approximated by a Gaussian distribution. Even though features 5 and 6 exhibit heavy skewness and sharp drop-offs, the unconstrained MVG is still able to extract enough discriminative information from the remaining features to achieve the lowest baseline error rate.

Apply now the tied Gaussian model, and compare the results with MVG and LDA. Which model seems to perform better?

The Tied Covariance MVG model results in an error rate of **9.3%**.

Compared to the Standard MVG (7.0%), the Tied model performs significantly worse. However, when we compare it to Linear Discriminant Analysis (LDA), the Tied MVG performs identically, as the LDA model also achieved an error rate of **9.3%**.

This identical performance makes perfect theoretical sense: both LDA and Tied MVG rely on the exact same fundamental mathematical assumption. They both assume that all classes share the same within-class covariance matrix, resulting in a strictly linear decision boundary.

The Standard MVG performs better than both because the underlying assumption of a shared covariance is false for this dataset. As we discovered in our exploratory analyses (Laboratories 2 and 4), the "Fake" and "Genuine" classes exhibit significantly different variances for several features (for example, Genuine Feature 1 has a much higher variance than Fake Feature 1). By forcing a single, pooled covariance matrix, the Tied MVG and LDA destroy this crucial geometric distinction. The unconstrained MVG successfully exploits these variance differences using a quadratic decision boundary, making it the better-performing model.

Finally, test the Naive Bayes Gaussian model. How does it compare with the previous two?

The Naive Bayes model achieves an error rate of **7.2%**.

It performs remarkably well—vastly outperforming the Tied MVG and LDA (9.3%) and coming extremely close to the fully parameterized Standard MVG (7.0%).

The Naive Bayes model simplifies the standard MVG by assuming that all features are conditionally independent within each class, meaning it mathematically forces all out-of-diagonal elements in the covariance matrices to zero. The fact that the error rate only degrades by a negligible 0.2% compared to the full MVG strongly implies that the features in this dataset are already largely uncorrelated. Because the cross-feature correlations do not carry significant discriminative weight, the Naive Bayes independence assumption

holds up well in practice. It easily surpasses the Tied model because it retains the class-specific variances that the Tied model discards, proving that capturing distinct variances is much more important for this dataset than capturing cross-feature correlations.

2 Covariance and correlation analysis

Start by printing the covariance matrix of each class. Compare the covariance of different feature pairs with the respective variances. What do you observe?

The empirical covariance matrices for the two classes, extracted directly from the Multivariate Gaussian model parameters, are printed below:

```
--- CLASS 0: Fake ---
Covariance Matrix:
[[ 0.6010  0.0001  0.0191  0.0193  0.0128 -0.0135]
 [ 0.0001  1.4472 -0.0161 -0.0159 -0.0265  0.0229]
 [ 0.0191 -0.0161  0.5653 -0.0018 -0.0069  0.0169]
 [ 0.0193 -0.0159 -0.0018  0.5416  0.0053  0.0136]
 [ 0.0128 -0.0265 -0.0069  0.0053  0.6961  0.0158]
 [-0.0135  0.0229  0.0169  0.0136  0.0158  0.6865]]

--- CLASS 1: Genuine ---
Covariance Matrix:
[[ 1.4481 -0.0147  0.0056  0.0157  0.0195 -0.0002]
 [-0.0147  0.5534 -0.0112 -0.0091 -0.0147  0.0163]
 [ 0.0056 -0.0112  0.5575  0.0276 -0.0038 -0.0146]
 [ 0.0157 -0.0091  0.0276  0.5697 -0.0117  0.0350]
 [ 0.0195 -0.0147 -0.0038 -0.0117  1.3420  0.0169]
 [-0.0002  0.0163 -0.0146  0.0350  0.0169  1.3037]]
```

Comparing the off-diagonal entries (covariances) with the main diagonal elements (variances), we observe that the variances are orders of magnitude larger than the covariances across both classes. For example, in the "Fake" class, the variance of Feature 2 is 1.4472, whereas its covariance with Feature 1 is a negligible 0.0001. Even the largest off-diagonal value in the dataset (the covariance between Feature 4 and Feature 6 in the Genuine class) reaches only 0.0350, which fades in comparison to their respective variances of 0.5697 and 1.3037. This pattern demonstrates that the directional spread of the data is concentrated along the primary axes of the feature space, indicating that cross-feature interactions are virtually non-existent.

Compute the correlation matrices for the two classes. Are the features strongly or weakly correlated? How is this related to the Naive Bayes results?

To evaluate the normalized strength of these relationships, the Pearson correlation matrices were computed using the formula $\text{Corr}_{i,j} = \frac{\text{Cov}(i,j)}{\sqrt{\text{Var}(i)\text{Var}(j)}}$:

```
--- CLASS 0: Fake ---
Correlation Matrix:
[[ 1.0000  0.0001  0.0327  0.0337  0.0198 -0.0210]
 [ 0.0001  1.0000 -0.0178 -0.0179 -0.0264  0.0230]
 [ 0.0327 -0.0178  1.0000 -0.0033 -0.0110  0.0271]
 [ 0.0337 -0.0179 -0.0033  1.0000  0.0086  0.0223]
```

```
 [ 0.0198 -0.0264 -0.0110  0.0086  1.0000  0.0229]
 [-0.0210  0.0230  0.0271  0.0223  0.0229  1.0000]]

--- CLASS 1: Genuine ---
Correlation Matrix:
[[ 1.0000 -0.0164  0.0062  0.0173  0.0140 -0.0001]
 [-0.0164  1.0000 -0.0202 -0.0161 -0.0170  0.0192]
 [ 0.0062 -0.0202  1.0000  0.0489 -0.0044 -0.0171]
 [ 0.0173 -0.0161  0.0489  1.0000 -0.0134  0.0406]
 [ 0.0140 -0.0170 -0.0044 -0.0134  1.0000  0.0128]
 [-0.0001  0.0192 -0.0171  0.0406  0.0128  1.0000]]
```

Looking at the off-diagonal correlation values, we can conclude that the features are exceptionally **weakly correlated**. Almost all correlation coefficients hover tightly between -0.03 and $+0.05$. The maximum linear dependency found in the entire dataset occurs between Feature 3 and Feature 4 in the Genuine class, reaching a correlation coefficient of just 0.0489, which still represents a negligible relationship.

This complete absence of correlation provides a direct mathematical explanation for the remarkable success of the Naive Bayes Gaussian classifier. The Naive Bayes model operates under the strict assumption that all features are conditionally independent given the class label, which forces all out-of-diagonal elements of the estimated covariance matrices to zero.

Because the empirical data shows that these off-diagonal correlation values are already effectively zero, the simplification made by Naive Bayes introduces almost no estimation error or structural distortion. By omitting the estimation of these redundant covariance parameters, the Naive Bayes model eliminates background noise without sacrificing discriminative capacity. This is why the Naive Bayes model matches the unconstrained Multivariate Gaussian classifier almost perfectly, achieving a highly competitive validation error rate of 7.2% compared to the full MVG's 7.0%.

3 Feature ablation study

Repeat the classification using only feature 1 to 4. What do you obtain?

When restricting the feature space to the first four features (Features 1, 2, 3, and 4) and discarding the final two, the validation error rates for the three generative classifiers are as follows:

- **Standard MVG:** 8.0% (from 7.0% to 8.0%)
- **Naive Bayes:** 7.7% (from 7.2% to 7.7%)
- **Tied Covariance MVG:** 9.5% (from 9.3% to 9.5%)

What can we conclude on discarding the last two features?

Comparing these results with our full 6-dimensional baseline model, we observe a clear degradation in classification performance across all three architectures. Specifically, the error rate of the unconstrained Standard MVG increases from 7.0% to 8.0%, while the Naive Bayes model degrades from 7.2% to 7.7%. From this degradation, we can definitively conclude that discarding the last two features results in a net loss of discriminative information, proving that Features 5 and 6 provide valuable class-separating utility when evaluating the full feature vector.

Despite the inaccuracy of the assumption for these two features, are the Gaussian models still able to extract some useful information?

Yes, the Gaussian models are unequivocally capable of extracting useful, predictive information from Features 5 and 6. As demonstrated in Laboratory 4, the 1D Maximum Likelihood density estimation showed that Features 5 and 6 are heavily skewed to the left and exhibit sharp drop-offs, making them a poor fit for a symmetric Gaussian distribution.

However, the fact that including these two features improves the full MVG error rate by a absolute 1.0% (dropping from 8.0% to 7.0%) demonstrates a vital machine learning principle: a model can tolerate a significant statistical mismatch if the feature provides strong, independent boundary information. The structural information gained by mapping the data along these two dimensions far outweighs the approximation errors introduced by the Gaussian model's rigid symmetry.

Repeat the classification using only features 1-2 and 3-4. Which model is better in each case?

Slicing the dataset into isolated 2-dimensional feature pairs reveals a complete reversal of model superiority depending on the selected subspace:

- **Case 1 (Features 1–2 jointly):** The Standard MVG and Naive Bayes models vastly outperform the Tied model. Standard MVG achieves an error rate of **36.5%** and Naive Bayes reaches **36.3%**, whereas the Tied Covariance MVG model collapses entirely, resulting in an error rate of **49.4%**.

standard MVG: 36.5%
Naive Bayes: 36.3%
tied covariance MVG: 49.4%

- **Case 2 (Features 3–4 jointly):** The performance among the classifiers becomes nearly uniform, but the Tied Covariance MVG emerges as the superior model, achieving the lowest error rate at **9.4%**, slightly outperforming both the Standard MVG (9.5%) and Naive Bayes (9.5%).

standard MVG: 9.5%
Naive Bayes: 9.5%
tied covariance MVG: 9.4%

How is this related to the characteristics of the two classifiers?

This compelling behavior is directly tied to the geometric constraints governing the decision boundaries of the respective classifiers. The Standard MVG estimates separate, class-specific covariance matrices, which mathematically translates into a complex, quadratic decision boundary capable of enveloping a class with a larger spread.

Conversely, the Tied MVG forces a single, pooled within-class covariance matrix across both classes. This constraint restricts its decision boundary to a strictly linear hyperplane. The performance in each subspace depends entirely on whether the physical separation of the classes is driven by their differences in coordinate means or their differences in directional variance.

Is the tied model effective at all for the first two features? Why?

No, the Tied model is completely ineffective when restricted to the first two features, as its 49.4% error rate is practically identical to a random coin flip. This total failure is explained by our exploratory analysis from Laboratories 2 and 4. For Features 1 and 2, the empirical means of the "Fake" and "Genuine" classes are located extremely close to one another, centered near zero. The true separation between the classes is driven entirely by their variances: for Feature 1, the Genuine variance (1.4302) is more than double the Fake variance (0.5696), whereas for Feature 2, the pattern flips and the Fake variance (1.4209) dominates the Genuine variance (0.5783).

Because the Tied model operates under the strict assumption that both classes share the exact same variance, it mathematically strips away the class-specific spreads. Since the means are overlapping, forcing equal covariances completely removes the primary source of discriminative information available in that subspace, leaving the linear boundary unable to separate the data.

And the MVG? And for the second pair of features?

The Standard MVG is highly effective for the first pair of features because its quadratic decision boundary is specifically designed to separate classes based on differences in variance, allowing it to easily draw a boundary that isolates the tightly clustered "Fake" class from the widely dispersed "Genuine" class.

For the second pair of features (Features 3 and 4), the underlying data characteristics are completely reversed. As shown in our MLE logs, the classes share nearly identical variances for these features (e.g., Feature 3 variance is 0.5500 for Fake and 0.5489 for Genuine), but their means are strongly separated on opposite sides of the origin (Fake mean is -0.6809 and Genuine mean is $+0.6652$).

Because the true class variances in this subspace are actually equal, the underlying assumption of the Tied MVG model is a perfect reflection of physical reality. A simple linear hyperplane is the mathematically optimal boundary to separate these two parallel, equal-variance Gaussian globs. While the unconstrained Standard MVG is flexible enough

to approximate this linear boundary, it wastes parameters estimating two separate covariance matrices from the validation data. By estimating only a single, shared covariance matrix, the Tied MVG eliminates estimation noise and avoids overfitting, which explains why it achieves a superior, highly robust error rate of 9.4%.

4 PCA pre-processing and conclusion

Use PCA to reduce the dimensionality of the feature space, and apply the three classification approaches. What do you observe?

By preprocessing the dataset with Principal Component Analysis (PCA) across a varying number of principal dimensions ($m = 1, \dots, 6$), we obtain the validation error rates compiled in Table 1.

PCA Dimension (m)	Standard MVG (%)	Naive Bayes (%)	Tied MVG (%)
1	9.3	9.3	9.4
2	8.8	8.9	9.3
3	8.8	9.0	9.3
4	8.0	8.9	9.3
5	7.1	8.8	9.3
6	7.0	8.9	9.3
Raw Baseline (6D)	**7.0**	**7.2**	**9.3**

Table 1: Generative classifier error rates as a function of the number of retained PCA dimensions (m) compared against the unreduced 6D baseline parameters.

Evaluating the trends in the experimental data, we observe distinct behaviors for each classification architecture:

- **Standard MVG:** The error rate scales inversely with the number of dimensions, steadily dropping from 9.3% at $m = 1$ down to 7.0% at $m = 6$. Retaining all 6 principal components perfectly recovers the raw unconstrained baseline error rate.
- **Tied MVG:** The error rate remains completely invariant at 9.3% for all dimensions $m \geq 2$, matching its baseline. It experiences a minor degradation to 9.4% only when compressed down to a single dimension ($m = 1$).
- **Naive Bayes:** The classifier displays an irregular performance profile that is consistently and significantly worse than its raw baseline. Even when keeping all dimensions ($m = 6$), its error rate settles at 8.9%, representing a steep drop in accuracy compared to its unprojected performance of 7.2%.

Is PCA effective for this dataset with the Gaussian models?

No, PCA is entirely ineffective as a predictive enhancement step for this specific dataset, and when paired with the Naive Bayes classifier, it proves highly destructive.

For the Standard and Tied MVG models, PCA fails to improve classification accuracy at any subspace dimension. Because PCA is an unsupervised technique, it blindly discards directions associated with the lowest global variance (S_T). The steady rise in the Standard MVG error rate as dimensions are discarded indicates that these low-variance components contain crucial discriminative info needed to form an accurate quadratic decision boundary.

The most vital finding, however, is why the Naive Bayes model degrades so heavily under PCA pre-processing (jumping from 7.2% to 8.9% error at $m = 6$). By definition,

PCA computes an orthogonal rotation that decorrelates the *entire dataset as a unified global pool* ($S_T = S_B + S_W$).

While this rotation successfully eliminates correlation across the global mixture, it can mathematically introduce significant cross-feature correlation *within the localized distributions of each individual class*. Because the Naive Bayes classifier operates on the strict structural assumption of conditional independence within each class, forcing the class-conditional covariance matrices to be strictly diagonal creates a severe mismatch against this newly rotated data geometry. The model tries to fit independent axis-aligned Gaussians to data clusters that have been diagonally elongated by the rotation, leading to a substantial increase in classification errors.

Overall, which is the model that provided the best accuracy on the validation set?

The model that provides the highest overall accuracy on the validation set is the **Standard Multivariate Gaussian (MVG) classifier operating on the full, unreduced 6-dimensional raw feature space**, achieving the optimal error rate of **7.0%**.

This unconstrained configuration succeeds where the other variations fail due to three distinct data characteristics:

1. Unlike the Tied MVG and LDA (9.3% error), the unconstrained MVG implements separate class covariance matrices, allowing its quadratic decision boundary to successfully exploit the massive differences in feature variance between the Fake and Genuine classes.

2. Unlike the Naive Bayes classifier post-PCA rotation, the raw full MVG retains the complete mathematical capacity to map minor cross-feature interactions without being disrupted by artificial class-conditional dependencies introduced via unsupervised coordinate transformations.

3. As established in our ablation study, although Features 5 and 6 violate the ideal symmetric Gaussian assumption due to left-sided skewness, they still contain essential boundary information. The unconstrained 6D MVG is robust enough to leverage this remaining information, outperforming its own 4-dimensional subspace alternative (8.0% error).

Therefore, preserving the full 6-dimensional unrotated feature space under a fully parameterized, class-specific Multivariate Gaussian distribution represents the mathematically optimal configuration for this pattern recognition task.